Note: Function prototype is mandatory since the function factorial() is given after the main() function.

11.10 Scope Rules of Variables

Scope refers to the accessibility of a variable. There are four types of scopes in C++. They are: **Local scope, Function scope, File scope** and **Class scope**.

11.10.1 Introduction

A scope is a region or life of the variable and broadly speaking there are four places, where variables can be declared,

- Inside a block which is called local variables.
- Inside a function is called function variables.
- Outside of all functions which is called global variables.
- Inside a class is called class variable or data members.

11.10.2 Local Scope:

- A local variable is defined within a block. A block of code begins and ends with curly braces { }.
- The scope of a local variable is the block in which it is defined.
- A local variable cannot be accessed from outside the block of its declaration.
- A local variable is created upon entry into its block and destroyed upon exit.

11.10.3 Function Scope:

- The scope of variables declared within a function is extended to the function block, and all sub-blocks therein.
- The life time of a function scope variable, is the life time of the function block. The scope of formal parameters is function scope.

11.10.4 File Scope:

- A variable declared above all blocks and functions (including main ( )) has the scope of a file. The life time of a file scope variable is the life time of a program.
- The file scope variable is also called as **global variable**.

Program 11.26

```
//Demo to test all Scopes//
#include<iostream>
using namespace std;
int file_var=20;          //Declared within File - file scope variable
void add(int x)
{
        int m;            //Declaration of  variable m in add () - Function scope variable
        m=x+30+file_var;
        cout<<"\n The Sum = "<<m;
}
```


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